

An ALUR Universality Reduction for Generating Operators

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Source Question

Kadets, Martin, Meri, and Quero ask in arXiv:2306.02645, after their Proposition 4.4, whether for a cardinal Γ there is a Banach space Y with $\text{dens}(Y) = \Gamma$ such that

$$\text{Gen}(X, Y) \neq \emptyset$$

for every Banach space X satisfying $\text{dens}(X) \leq \Gamma$. Here $\text{Gen}(X, Y)$ denotes the norm-one generating operators from X to Y .

Definitions Used

A Banach space is ALUR, or average locally uniformly rotund, when every point of its unit sphere is a denting point of its unit ball. The source paper records the standard equivalence and proves the following corollary (`cor:domain-LUR` in the source): if X is ALUR, then every generating operator $G: X \rightarrow Y$ is an isometric embedding.

Proof Intuition

The density-character question is phrased in terms of generating operators, which are a priori weaker than isometric embeddings. However, the weakening disappears on ALUR domains. On such spaces, the source paper's denting-point criterion forces a generating operator to preserve the norm at every unit vector, hence to be an isometry. Therefore any single range space Y that works for all domains of density at most Γ must already be isometrically universal for the ALUR members of that class.

Theorem 1 (ALUR universality reduction). *Let Γ be a cardinal and let Y be a Banach space such that $\text{Gen}(X, Y) \neq \emptyset$ for every Banach space X with $\text{dens}(X) \leq \Gamma$. Then every ALUR Banach space X with $\text{dens}(X) \leq \Gamma$ embeds linearly isometrically into Y .*

Proof. Fix an ALUR Banach space X with $\text{dens}(X) \leq \Gamma$. By the assumed property of Y , there is a generating operator $G \in \text{Gen}(X, Y)$. By Corollary `cor:domain-LUR` of the source paper, every generating operator whose domain is ALUR is an isometric embedding. Hence G is a linear isometry from X into Y . $\square$

Corollary 1. *If a Banach space Y of density Γ answers the source question positively, then Y contains an isometric copy of $\ell_2(\kappa)$ for every cardinal $\kappa \leq \Gamma$.*

Proof. Hilbert spaces are uniformly convex, hence ALUR, and $\text{dens}(\ell_2(\kappa)) = \kappa$ for infinite κ . $\square$

Verification Notes

The proof uses only the source paper’s own Corollary `cor:domain-LUR` and the exact density-character question stated after Proposition 4.4. No additional set-theoretic assumption is used.

This is a necessary condition, not a positive or negative solution of the source question. It says that any positive answer must be at least as strong as isometric universality on the ALUR subcategory, so non-isometric generating operators cannot bypass isometric universality for ALUR domains.

Novelty and Duplicate Check

The run’s lightweight indexes were searched for 2306.02645, `generating operators`, `Gen(X,Y)`, density-character wording, and universal-space keywords. The only exact hit was the prior attempt `2306.02645_density_character_generating_operators_no_promotion.md`. That attempt records the invalid universal compact-space shortcut and does not contain the ALUR universality reduction stated here.

Limitations

The result does not prove that such a range space Y exists, nor does it prove nonexistence in ZFC. It also does not reduce the question to isometric universality for all Banach spaces, only for the ALUR subclass.

Reference

V. Kadets, M. Martin, J. Meri, and A. Quero, *Generating operators between Banach spaces*, arXiv:2306.02645.